UC San Diego
RADY SCHOOL OF MANAGEMENT

# MGTA 466
# Scalable Analytics
## WINTER 2026

| | |
|---|---|
| PROFESSOR: | Mai H. Nguyen |
| EMAIL | mhnguyen@ucsd.edu |
| OFFICE HOURS | TBD |
| | |
| TEACHING ASSISTANT | TBD |
| EMAIL | |
| OFFICE HOURS | |

### DESCRIPTION

Analyzing large-scale and complex data is an integral part of many business applications today. This course is designed to provide students with skills and knowledge to perform analytics at scale. Topics include steps in the analytics process used to gather, prepare, process, and evaluate data; basic principles of computer systems and parallelism; scalable computing; and cloud-based analytics. Students will be introduced to tools and techniques used to perform analytics on big data. Hands-on experience with widely used scalable and cloud platforms will be provided. Through programming assignments, students will apply what they learn in the course to leverage big data.

### OBJECTIVES

At the close of this course, you will be able to:
- Use scalable analytics algorithms and distributed platforms to leverage big data
- Analyze large-scale and complex data in real-world business applications
- Upload data and create analytics models using cloud-based analytics tools
- Create an analytics solution to address a business problem with big data
- Apply the analytics process to prepare, process, and interpret data
- Understand the basic principles of computer systems and parallelism

### PREREQUISITE

Since all programming assignments are in PySpark and Python, solid Python programming experience is required.

### MATERIALS

Required
- The Data Scientist's Guide to Apache Spark (PDF)
- Apache Spark
  - https://spark.apache.org/docs/latest/
- Amazon EMR
  - https://docs.aws.amazon.com/emr/latest/ManagementGuide/emr-what-is-emr.html
  - https://docs.aws.amazon.com/emr/

- Amazon SageMaker
  - https://docs.aws.amazon.com/sagemaker/latest/dg/how-it-works.html
  - https://docs.aws.amazon.com/sagemaker/latest/dg/gs.html

Recommended
- Spark: The Definitive Guide
- Learning Spark

**SCHEDULE**

| Session | Topic | Subtopics | Assignment | Points |
|---|---|---|---|---|
| Session 1 | Computer Systems & Parallelism | Basics of computer hardware and software<br>Memory hierarchy<br>Parallelism principles<br>Big Data characteristics and challenges | PA1 (Spark) | 10 |
| Session 2 | Big Data & Distributed Processing | Scalable systems<br>Hadoop<br>Spark | PA2 (Spark)<br>Quiz | 15<br>5 |
| Session 3 | Big Data Analytics | Analytics process<br>Spark core<br>Spark libraries<br>Analytics with Spark MLlib<br>PySpark, SparkR | PA3 (Spark)<br>Quiz | 20<br>5 |
| Session 4 | Cloud Analytics | Introduction to cloud computing<br>AWS Basics<br>Amazon EMR | PA4 (AWS)<br>Quiz | 15<br>5 |
| Session 5 | AWS Analytics | Model evaluation<br>Amazon SageMaker | PA5 (AWS)<br>Quiz | 20<br>5 |

**GRADING**

Grades are based on weekly programming assignments and several quizzes, with the following point breakdown:
- Assignments: 80%
- Quizzes 20%

Notes about Assignments:
- Programming assignments are assigned every week.
- PySpark and Python will be used for the programming assignments.
- Assignments are to be individual work only.
- Generative AI tools such as ChatGPT are to be used judiciously and must be well-documented. Proper use of these tools will be discussed in class.

Notes about Quizzes:
- Quizzes will be taken in class.
- To ensure fairness and integrity, a strict no-materials policy is enforced for the quizzes. This means that notes, course materials, the Internet, generative AI tools, etc. cannot be used during quizzes.

UC San Diego
RADY SCHOOL OF MANAGEMENT

## LATE POLICY

A late penalty of 10% per day will be applied if an assignment is submitted after the due date. A late submission can be accepted up to 3 days after the due date.

There is no make-up for the quizzes. Please plan accordingly.

## COURSE POLICIES

Students are encouraged to attend all class sessions and participate in the discussions to maximize their learning opportunities. Participation to promote understanding and thoughtful discussion of the course material is encouraged

## ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2

## STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the OSD website for further information or contact the Office for Students with Disabilities at (858) 534-4382 or **osd@ucsd.edu.**